

Han Bao

✉ Email | 🌐 Website | 🐙 Github | 🎓 Google Scholar

Education

- **University of Notre Dame** Sep 2026 – June 2031(Expected)
Ph.D in Computer Science, supervised by Prof. Fanny Ye. South Bend, USA
- **Sichuan University** Sep 2021 – June 2026
Bachelor of Engineering in Cyber Science and Engineering Chengdu, China
 - GPA: 3.56 / 4.00

Research Experience

- **National University of Singapore** 🌐 July 2024
Summer Workshop Participant Singapore
 - **Advisor:** Prof. Tianbai Ma
 - **Research Area:** Cloud Computing, Network Traffic Analysis
- **MINE Lab, University of Notre Dame** 🌐 Aug 2024 – Present
Visiting Research Student South Bend, USA
 - **Advisor:** Prof. Xiangliang Zhang
 - **Research Area:** Trustworthy Generative Models, LLM Alignment.

Publications

C=Conference, J=Journal, S=Submission/Preprint

- [J.1] **CRISP: A Causal Relationships-Guided Deep Learning Framework for Advanced ICU Mortality Prediction**
Linna Wang, Xinyu Guo, **Han Bao**, Lihua Jiang, Li Zhao, Tao Zhu, Ziliang Feng
Accepted by BMC Medical Informatics and Decision Making.
- [C.1] **LiD-FL: Towards List-Decodable Federated Learning**
Hong Liu, Liren Shan, **Han Bao**, Ronghui You, Yuhao Yi, Jiancheng Lv
Accepted by AAAI Conference on Artificial Intelligence (AAAI 2025).
- [C.2] **On the Trustworthiness of Generative Foundation Models: Guideline, Assessment, and Perspective**
Yue Huang, Chujie Gao, Siyuan Wu, ..., **Han Bao**, et al.
Accepted by ICLR 2026.
Project page: trustgen.github.io. *Role: Leader of T2I Group.*
- [C.3] **IntraAI: Bridging Human-AI Understanding Through Intelligent Interface Design**
Han Bao, Yue Huang, Xiangliang Zhang, Yanfang Ye
Accepted by WWW 2026.
Project page: [Intra_AI](https://intra-ai.github.io)
- [S.1] **AutoDavis: Automatic Benchmarking of Large Vision-Language Models on Visual Question-Answering**
Han Bao*, Yue Huang*, Yanbo Wang, Jiayi Ye, Xiangqi Wang, Xiuying Chen, Mohamed Elhoseiny, Xiangliang Zhang
Preprint on arXiv. (*Equal Contribution)
- [S.2] **PolicyLLM: Towards Excellent Comprehension of Public Policy for Large Language Models**
Han Bao, Penghao Zhang, Yue Huang, Rui Su, Zhengqing Yuan, Yujun Zhou, Xiangqi Wang, Kehan Guo, Nitesh Chawla, Xiangliang Zhang
Manuscript in preparation.
- [S.3] **Capability-Oriented Training Induced Alignment Risk**
Yujun Zhou*, **Han Bao***, Yue Huang, Kehan Guo, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Werner Geyer, Nuno Moniz, Nitesh V Chawla, Xiangliang Zhang
- [S.4] **Guardian-as-an-Advisor: Advancing Next-Generation Guardian Models for Trustworthy LLMs**
Yue Huang, Haomin Zhuang, Jiayi Ye, **Han Bao**, Yanbo Wang, Hang Hua, Siyuan Wu, Pin-Yu Chen, Xiangliang Zhang

- [S.5] Drift-Bench: Diagnosing Cooperative Breakdowns in LLM Agents under Input Faults via Multi-Turn Interaction**
Han Bao, Zheyuan Zhang, Pengcheng Jing, Zhengqing Yuan, Kaiwen Shi, Yanfang Ye
- [S.6] AI Alignment Breaks at the Edge**
Han Bao, Yue Huang, Xiaoda Wang, Zheyuan Zhang, Yujun Zhou, Carl Yang, Xiangliang Zhang, Yanfang Ye
- [S.7] SKILLGEN: Verified Inference-Time Agent Skill Synthesis**
Yuchen Ma, Yue Huang, Han Bao, Haomin Zhuang, Swadheen Shukla, Michel Galley, Xiangliang Zhang, Stefan Feuerriegel
- [S.8] Confidence Laundering in Agent Systems: Why Uncertainty Needs a Latent Carrier**
Kaiwen Shi, Zheyuan Zhang, Han Bao, Colby Nelson, Yanfang Ye
- [S.9] UXBench: Measuring the Actionability of LLM-Generated UX Critiques**
Wenjie Wang, Yue Huang, Zipeng Ling, Han Bao, Hang Hua, Xiaonan Luo, Yu Jiang, Shiyi Du, Yuexing Hao, Xiaomin Li, Yuchen Ma, Dianzhuo Wang, Yanfang Ye, Xiangliang Zhang
- [S.10] SpecAlign: Efficient Specification-Grounded Alignment of Large Language Models via Synthetic Data**
Wenjie Wang, Yue Huang, Zhengqing Yuan, Han Bao, Shiyi Du, Yuchen Ma, Yue Zhao, Yanfang Ye, Xiangliang Zhang
- [S.11] Why Semantic Entropy Fails? Geometry-Aware and Calibrated Uncertainty for Policy Optimization**
Zheyuan Zhang, Kaiwen Shi, Han Bao, Zehong Wang, Tianyi Ma, Yanfang Ye

Selected Projects

- **Web-based Advanced Medical Image Processing Platform** Oct 2023 – Sep 2024
Tools: Segment Anything Model (SAM), Glance, JavaScript, WebAssembly
 - Integrated the Segment Anything Model (SAM) for interactive and precise medical image segmentation.
 - Optimized the rendering pipeline by replacing the traditional OHIF viewer with the Glance viewer, significantly improving loading speeds.

Honors and Awards

- **Provincial Third Prize – China Undergraduate Mathematical Contest in Modeling** Nov 2023
Chinese Society for Industrial and Applied Mathematics (CSIAM)
 - Recognized for excellence in mathematical modeling and collaborative problem-solving under tight deadlines.
- **Second-Class Scholarship (Top 10%)** Sep 2023
School of Cyber Science and Engineering, Sichuan University
 - Awarded for outstanding academic achievement and research potential.

Technical Skills

- **AI & Machine Learning:** PyTorch, Transformers, LLaMA-Factory, LoRA/QLoRA, Reinforcement Learning (RL, VERL framework), Agentic Workflows
- **Development:** Python, C/C++, Docker, Git, MySQL, Redis, Wireshark, HTML/CSS
- **Research:** Experimental Design, Technical Writing, Data Visualization, Literature Review
- **Languages:** English (IELTS 7.0), Mandarin Chinese (Native)

Academic Service

- **Program Committee / Reviewer** Nov 2025
NeurIPS 2025 Workshop on Socially Responsible and Trustworthy Foundation Models